Introduction to Missing Data

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Overview

- Developing intuition about selection and selection bias
  - collider stratification
  - selection bias without collider stratification
- Missing data concepts
- Missing data handling methods
  - Complete case analysis
  - Ad Hoc Procedures
  - Weighting Procedures
  - Multiple Imputation

Developing data intuition

- studying human beings is complicated
  - rarely are we in a position to do tightly controlled experiments
  - especially in observational studies, we have to collect and analyze data on multiple variables to understand a phenomenon well
- we need to develop good intuition about biases and threats to the validity of our inferences

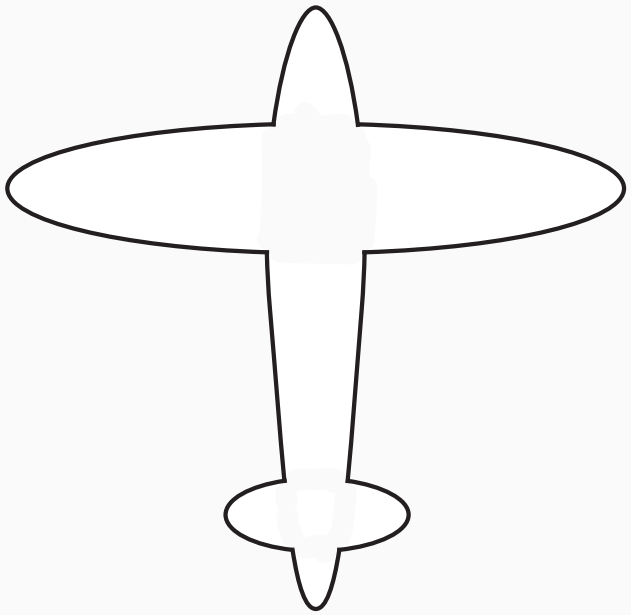
Developing intuition about selection

Selection and selection bias

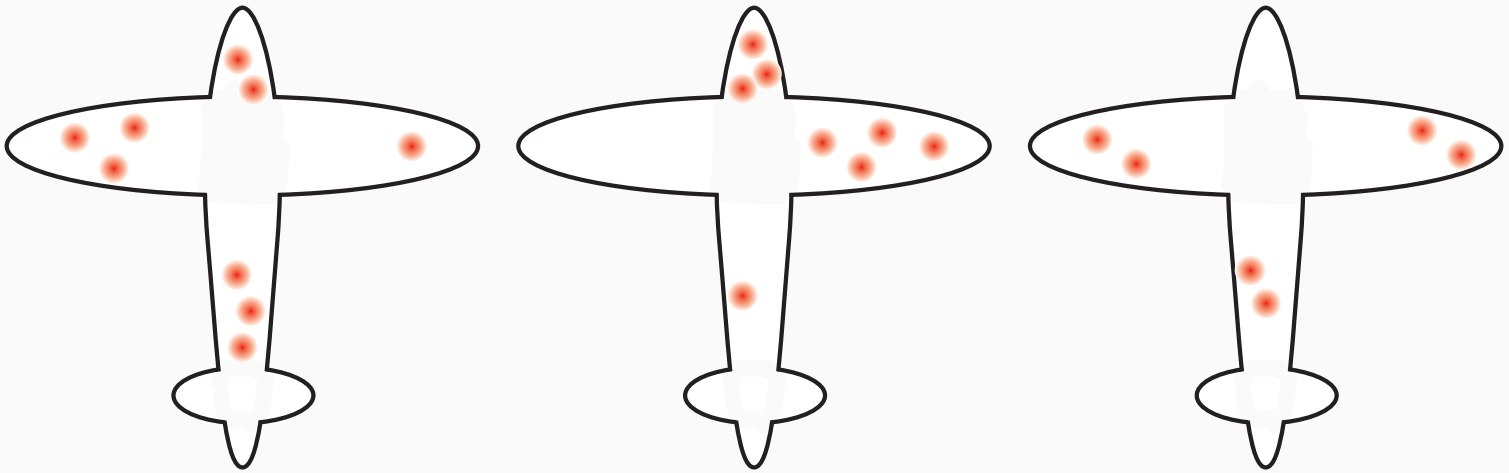
Selection can occur when the study sample we have observed is not representative of the target population from which it was drawn. This *can* lead to bias in estimates of causal effects.

- it's useful to distinguish **selection** from **selection bias**
  - **selection** refers to the non-representativeness of the observed cases available for analysis
  - as we discussed in PHS2000A, **bias** needs to be considered in relation to whether a particular estimator provides a consistent estimate of an estimand of interest
- Note that there are differences in how economists and epidemiologists use the term **"selection"**
  - "selection into treatment"
  - "selection into the study population"

An intuitive example of selection bias



An intuitive example of selection bias



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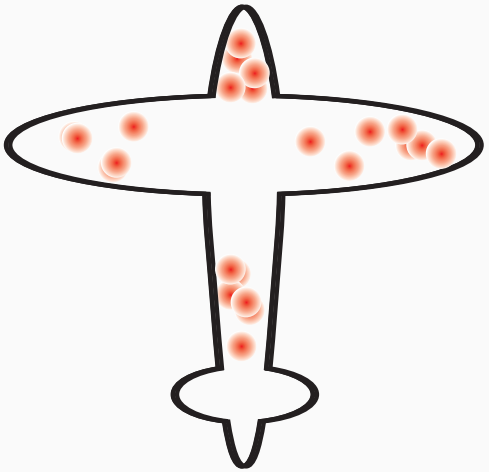

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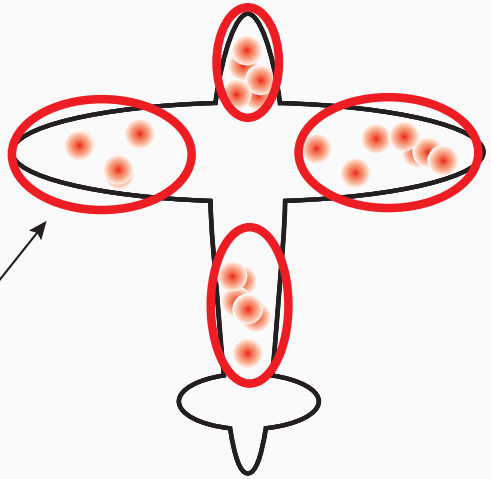
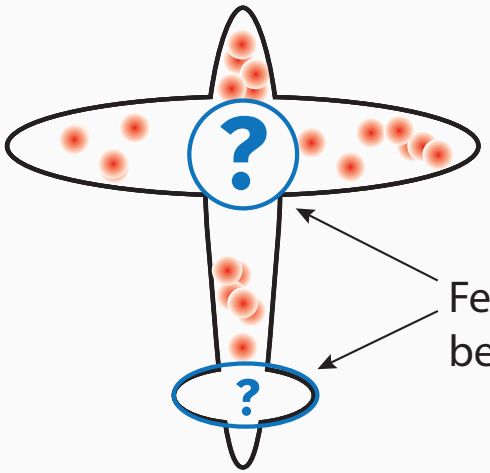


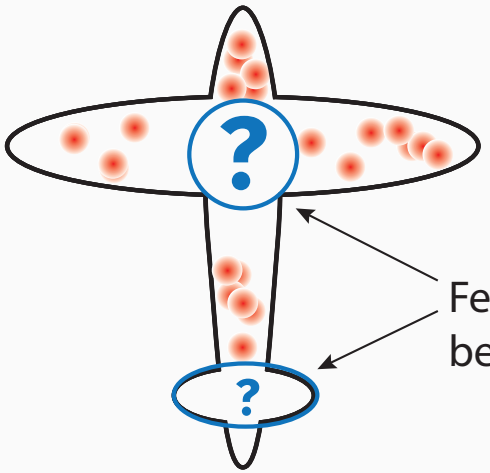
Initial conclusion:
reinforce the armor in
these most frequently
hit areas

An intuitive example of selection bias



Few returning planes had been hit in these areas.

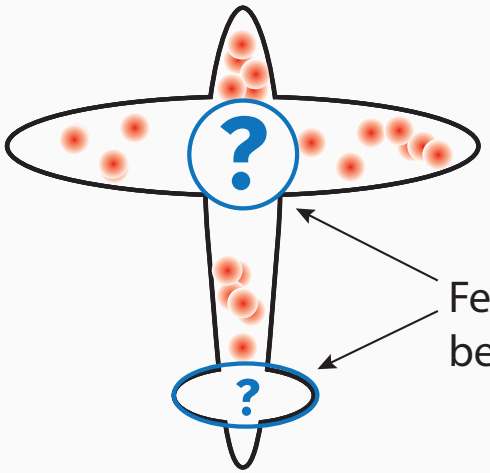
An intuitive example of selection bias



Few returning planes had been hit in these areas.

...but what about planes that did not return?

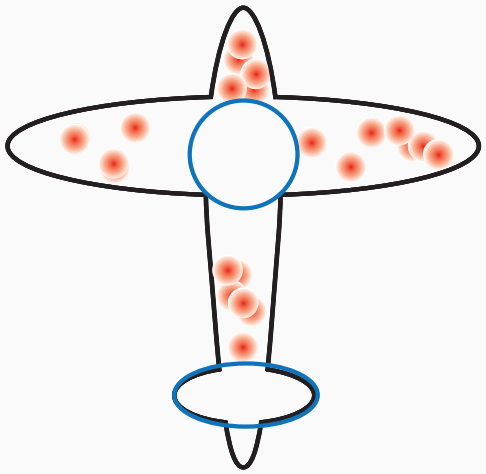
An intuitive example of selection bias



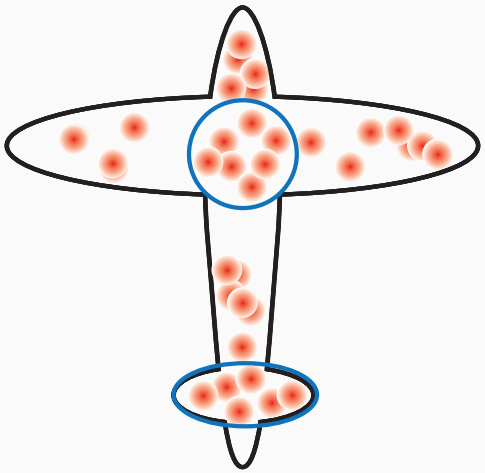
Few returning planes had been hit in these areas.

Abraham Wald recommended reinforcing these areas.
The British RAF followed suit and markedly fewer casualties were observed after the changes were made.

An intuitive example of selection bias



Observed



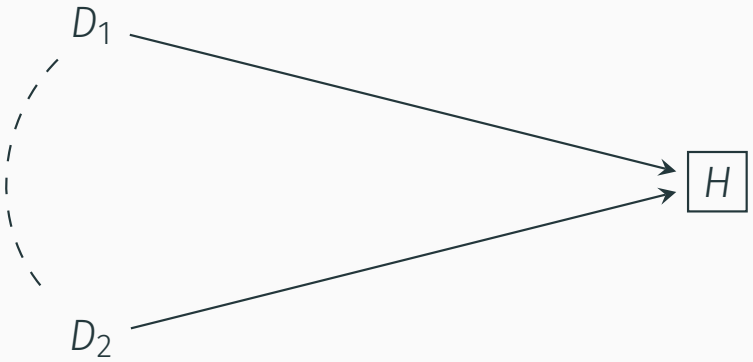
Unobserved

Key intuition: the wrong conclusion was made because observation was restricted only to planes that made it back.

The distribution of bullet holes on plane bodies is uniform, but because hits to the tail or the center part of the plane are catastrophic, planes that are observed tend not to have bullet holes in these places.

Berkson's fallacy

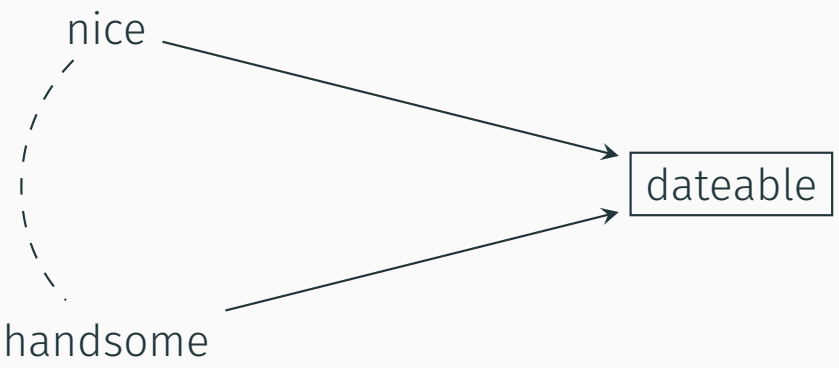
Berkson's fallacy: when two diseases, D_1 and D_2, that lead to hospitalization (H) are observed *among hospital patients*, they appear correlated even though they are independent.



Berkson's original example: is diabetes (D_1) a risk factor for cholecystitis (D_2) in a sample from a hospital in-patient population? (What direction do you expect the association to go?)

Are handsome men more likely to be rude?

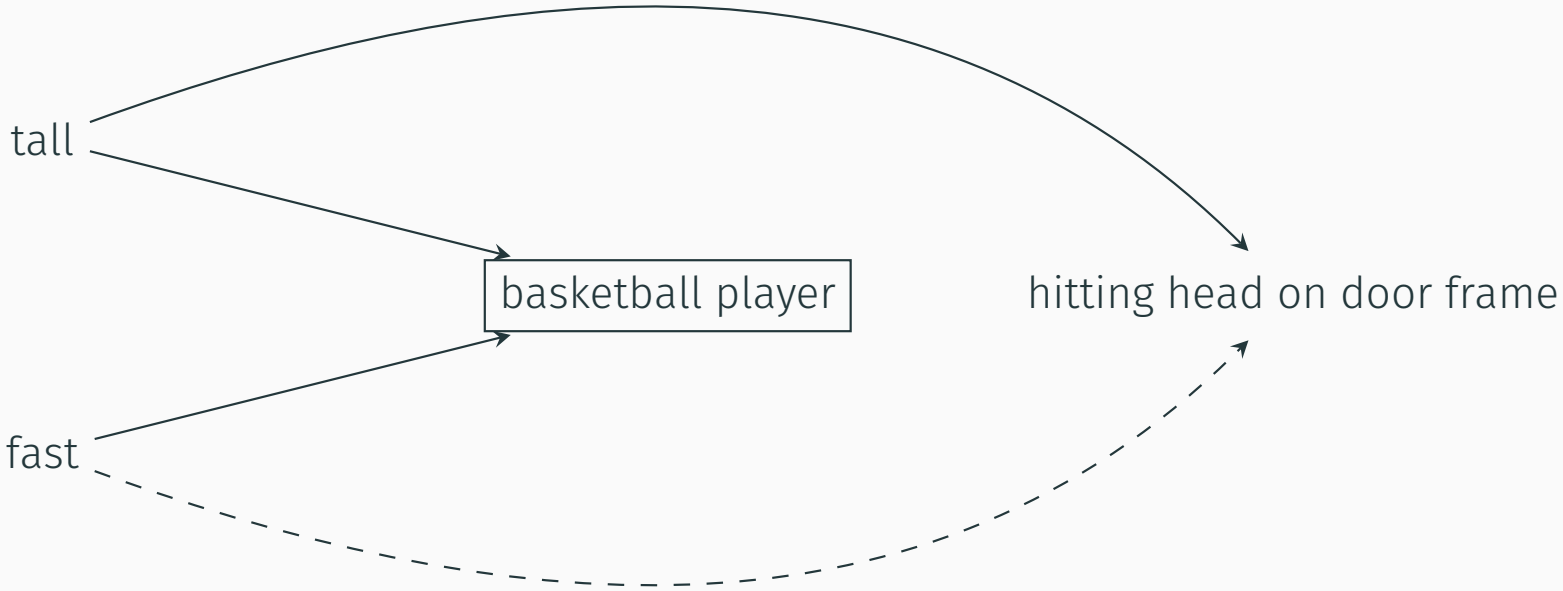
Suppose that Jarvis will only date a man if his niceness plus his handsomeness exceeds some threshold. That is, niceness and handsomeness are both predictors of being in Jarvis' dating pool.



Among the men that Jarvis dates, Jarvis may observe that the nicer ones are less handsome on average and vice versa, even if these traits are uncorrelated in the general population. [1]

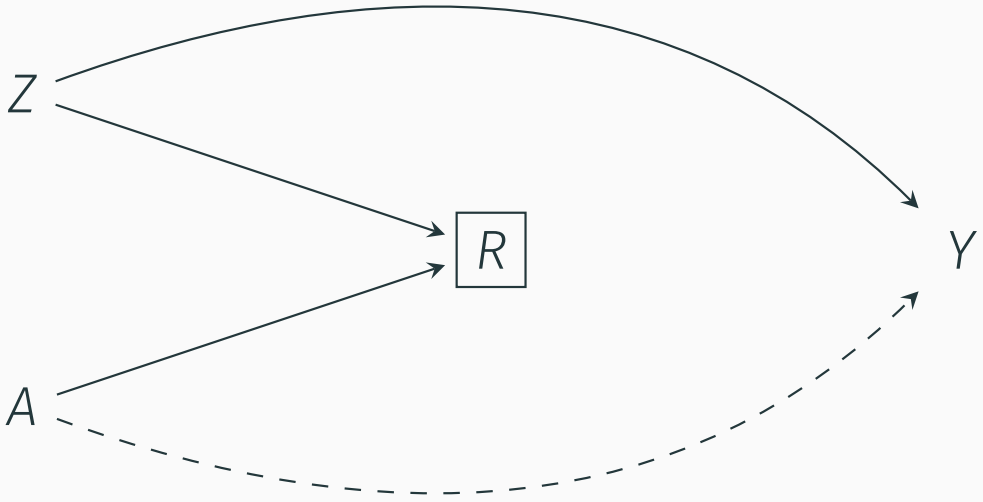
[1] Example attributed to the mathematician Jordan Ellenberg

Another example



- Being tall and being fast are associated with being a basketball player.
- Given that you are not tall, if you are a basketball player, you must be fast.
- Among basketball players, being fast appears to be protective against hitting your head on doorframes.

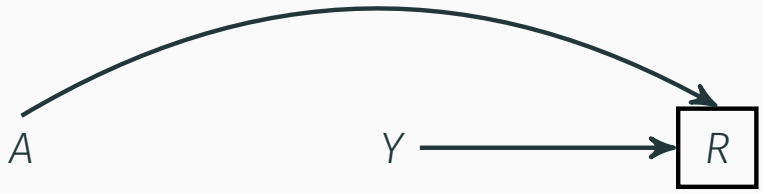
Collider stratification



- **Selection** shares this general structure of *conditioning* on a common effect (here, restricting observation to the stratum where $R = 1$).
- Note that this is not a situation where we have "inappropriately" conditioned on a collider in the analysis! Rather, restriction to $R = 1$ in the **design** of the study has induced collider stratification bias (the data came to us this way!)

Structural Approaches to Selection Bias

More recent literature has focused on **structural representations of missing data** (and also of **selection bias** more broadly).[2] For example, some of the earliest causal DAGs you saw probably depicted a simple selection bias scenario resulting from **collider stratification:**



Where A is the exposure, Y is the outcome, and R is an indicator for selection into the study.

[2] Hernán MA, Hernández-Díaz S, Robins JM. A structural approach to selection bias. Epidemiology. 2004 Sep;15(5):615-25

Revisiting Selection Bias

In some settings, bias due to conditioning on having observed (non-missing) data may be represented by collider stratification on a causal DAG. However, bias due to missing data may also be present even in the absence of collider stratification! To understand why this is the case, we will start by providing a very general definition of selection bias:

Selection bias: The parameter of interest in the target population differs from the parameter in the subset of individuals from the population available for analysis ("study sample").[3]

[3] Many epidemiologists use the term selection bias to refer exclusively to biases that affect internal validity. The definition proposed here encompasses biases which can result from either a lack of internal or external validity. This approach is aligned with a recent body of literature on *target validity*.

A Note on Selection Bias vs. Selection

Structural representations can also help us distinguish between selection vs. selection bias:

Selection:
The observed study sample is not representative of the target population of interest. Whether or not bias is induced depends on the estimand we wish to estimate.

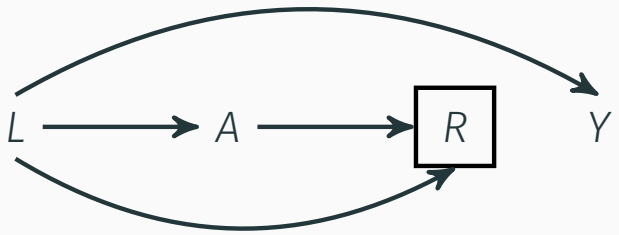
Selection Bias
The target estimand of interest is not equal to the estimate obtained from the observed data as a result of selection. Results in biased estimates.

Structural Representations of Selection Bias with Collider Stratification

We can now differentiate **two potential sources of selection bias.**

1. **Selection bias due to collider stratification:** Conditioning on a collider (or a descendent of a collider) on a path between exposure A and outcome Y generally induces a non-causal $A - Y$ association, even if the exposure has no causal effect on the outcome (i.e., *selection bias under the null*).[4]

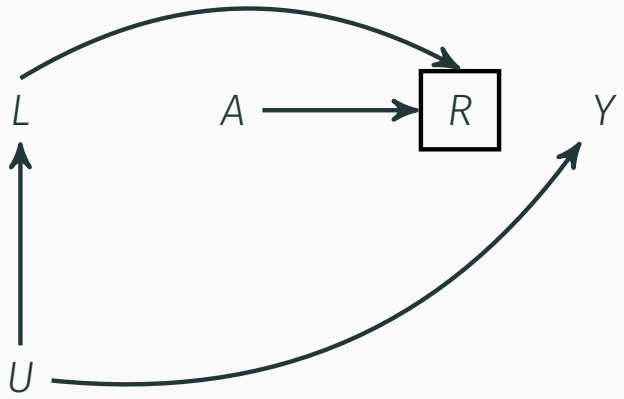
   For example, consider this DAG where R represents missing data and L is a confounder:



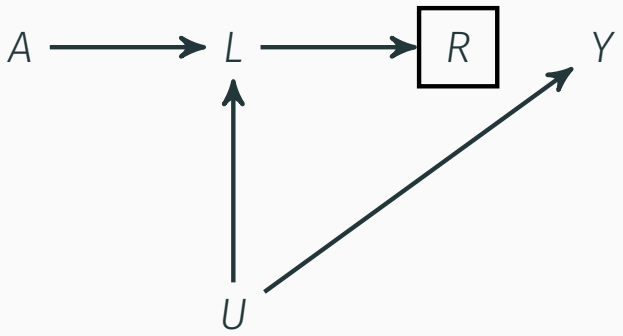
[4] Although collider stratification always induces an association between the causes of the collider, that association may in some cases be restricted to specific levels of the collider.

Structural Representations of Selection Bias with Collider Stratification

a) Selection bias is induced through the $A \leftarrow R \rightarrow L \leftarrow U \rightarrow Y$ path since we are conditioning on R, which is a collider.



b) Selection bias is induced through the $A \leftarrow L \leftarrow U \rightarrow Y$ path since we are conditioning on R, which is a descendant of a collider, L.



Structural Representations of Selection Bias without Collider Stratification

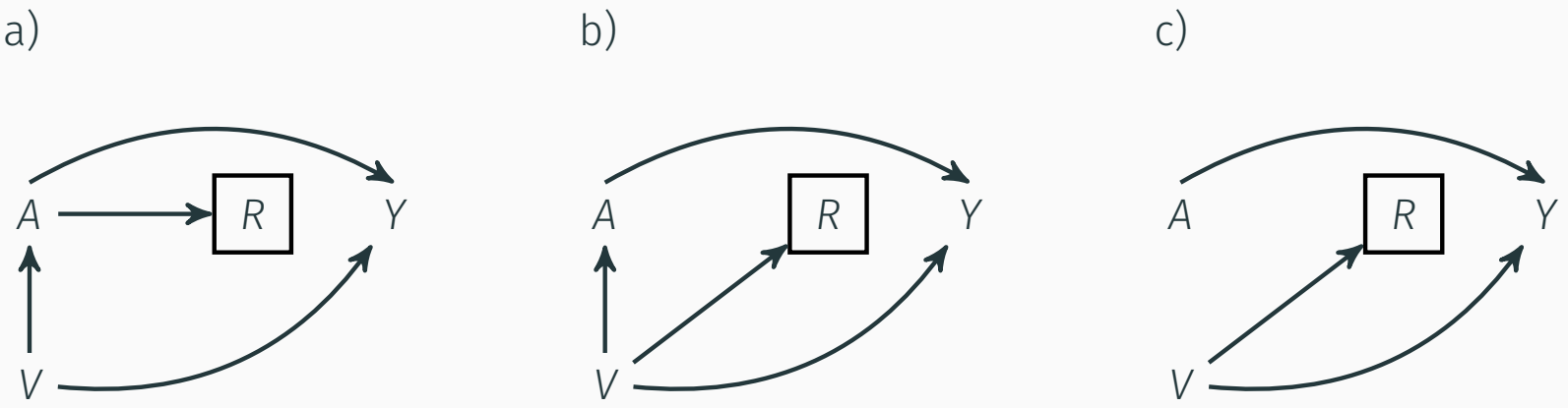
Under certain conditions, selection bias *may occur in the absence of colliders*!

2. **Selection bias without colliders:** Selection occurs when the observed study sample is not equal to the target population of interest. When selection is present but there are no colliders, the estimated quantity may be unbiased for the subset of the population that was selected but *biased in the entire target population*. This form of selection bias essentially occurs when the selected sample is not representative of the target population $\implies$ **failure to generalize.** This concept applies to both descriptive and causal measures.[5]

[5] Greenland S. Response and follow-up bias in cohort studies. Am J Epidemiol. 1977 Sep;106(3):184-7.,[6]

Selection Bias without Collider Stratification: Causal Effects

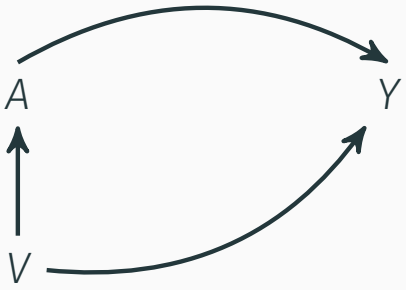
Selection bias for a causal effect without colliders: Selection bias may arise in the absence of colliders when, on the scale of interest for the analysis, the *effect of the exposure on the outcome is heterogeneous across some level(s) of V*, where *V* represents a third variable that is associated with selection. Such bias only arises when the *exposure has a non-null effect on the outcome.* For example [7]



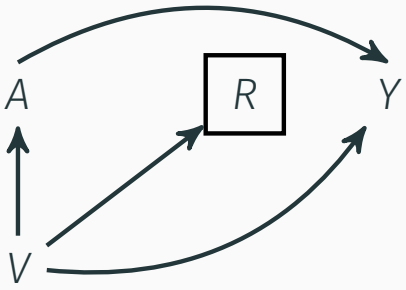
[7] These DAGs were adapted from Smith, L.H. Selection Mechanisms and Their Consequences: Understanding and Addressing Selection Bias. Curr Epidemiol Rep 7, 179–189 (2020).

Selection Bias without Collider Stratification: Causal Effects

DAG A: No Selection

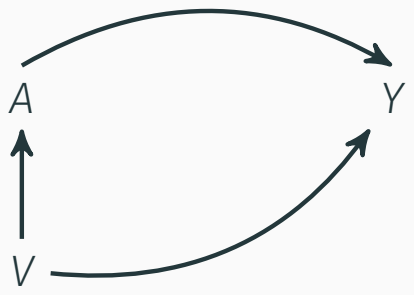


DAG A: Selection

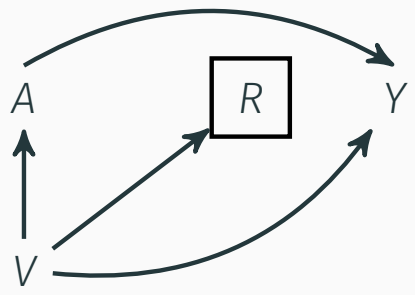


Selection Bias without Collider Stratification: Causal Effects

DAG A: No Selection



DAG A: Selection



- Even though $\mathbb{E}[Y \mid A = a, V = v] = \mathbb{E}[Y \mid A = a, V = v, R = 1]$ for both exposure values, we expect the distribution of V to differ in the selected sample compared with the target population, since V influences R: $V \not\perp\!\!\!\perp R$ and $Pr[V = v] \neq Pr[V = v \mid R = 1]$.
- Suppose we are interested in estimating the causal difference

$$\sum_v \Big\{\mathbb{E}(Y \mid A = a, V = v, R = 1) - \mathbb{E}(Y \mid A = a^*, V = v, R = 1)\Big\} Pr(V = v, R = 1)$$

  This will not be equal to

$$\sum_v \Big\{\mathbb{E}(Y \mid A = a, V = v) - \mathbb{E}(Y \mid A = a^*, V = v)\Big\} Pr(V = v)$$

  if there is heterogeneity in $\mathbb{E}(Y \mid A = a, V = v) - \mathbb{E}(Y \mid A = a^*, V = v)$ across levels of V.
- That is, unless the causal effect on the scale of interest is homogeneous across values of V, the average effect in the target population will differ from the effect in the selected study sample.

Missing data concepts

Missing Data

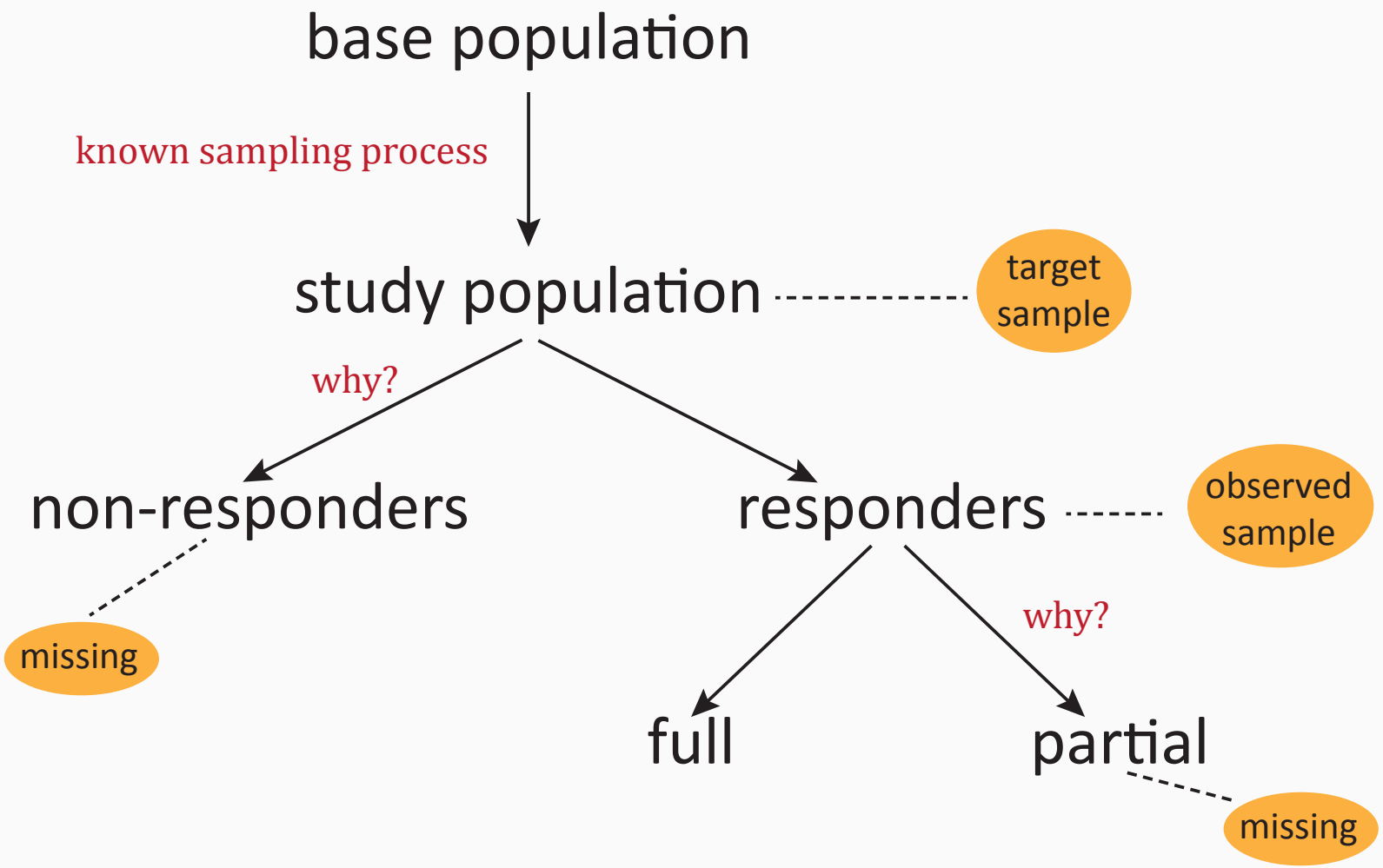
In many studies, measurements of every variable are not available for every subject. Data may be missing by chance or by design.

- unavailability of covariate measurements
- survey non-response
- some study subjects fail
  to report to a clinic for monthly evaluations
- respondents refuse
  to answer certain items on a questionnaire
- loss of data
- some information may be suppressed
  on purpose, e.g. to protect confidentiality

Missing data are common

- Often inadequately handled in both observational and experimental research.
- For example, Wood et al. (2004) reviewed 71 recently published BMJ, JAMA, Lancet, and NEJM papers.
  - 89% had partly missing outcome data.
  - In 37 trials with repeated outcome measures, 46% performed complete case analysis.
  - Only 21% reported sensitivity analysis
- Sterne et al. (2009) reviewed articles using Multiple Imputation in BMJ, JAMA, Lancet, and NEJM from 2002 to 2007.
  - 59 articles found, with use doubling over 6 year period
  - However, the reporting was almost always inadequate.

How do missing data arise?



Missing data in longitudinal settings

Time →

	t_1	t_2	t_3	t_4	t_5	t_6	
Subject 1	y_1	y_2	y_3	y_4	y_5	y_6	*completely observed*
Subject 2	y_1	y_2	y_3	*	*	*	*lost to follow up at t_4*
Subject 3	y_1	*	y_3	y_4	*	y_6	*intermittent missingness*
Subject 4	y_1	*	*	y_4	*	*	*measurements at t_1 and t_4 by design*
Subject 5	*	*	*	y_4	y_5	y_6	*outcome not defined until t_4*

** denotes missing value*

Consequences of missing data

- Intuitively, when the subjects with missing covariate values differ systematically from those with complete data with respect to the outcome of interest, results from a traditional data analysis omitting the missing cases may no longer be valid.
- If subjects with any missing covariates are excluded from the analysis, analyses can be **biased** and **inefficient**.

Classification of Missing Data

In their canonical work on missing data, Little and Rubin[8] classified missing data into three categories:

- Missing Completely At Random (MCAR)
- Missing At Random (MAR)
- Missing Not At Random (MNAR or NMAR)

[8] Rubin DB (1976). Inference and missing data. Biometrika 63:581-592.,[9]

Classification of Missing Data

- Let's assume that data are collected on a sample of n subjects and that primary interest relates to the parameters governing the conditional distribution, $f(Y_i|\mathbf{X}_i, \boldsymbol{\beta})$. To simply exposition, we suppress the subject indicator i.
- For full generality, we'll use $\mathbf{Z} = (Y, \mathbf{X})$ to denote the set of all variables of interest, including outcomes (Y) and predictors ($\mathbf{X}$).
- For a given subject, we can partition $\mathbf{Z}$ into components denoting observed variables ($\mathbf{Z}^{obs}$) and those that are missing for that subject ($\mathbf{Z}^{mis}$).
- We denote by $\mathbf{R}$ a set of response indicators where $R_j = 1$ if the jth element of $\mathbf{Z}$ is observed and equals 0 otherwise.
- Little and Rubin proposed a classification for missingness in terms of probability models for $\mathbf{R}$.

Notation

Notation	Definition
Z	A vector of all variables of interest in a given analysis, including the outcome, exposure(s), and all covariates
Z^{obs}	A (sub) vector of variables within Z which are **observed** for a given individual i
Z^{mis}	A (sub) vector of variables within Z which are **missing** (unobserved) for a given individual i
R	A missing indicator vector which has elements that take the value 1 for all Z^{obs} (observed covariates) and 0 for all Z^{mis} (missing covariates)

Example: We are interested in analyzing the vector of variables $Z = \{$ age, highest level of education, earnings $\}$. For an arbitrary individual i, let $Z_i = \{30, NA, NA\}$ where NA indicates a missing value. Thus, for this individual i, $R_i = \{1, 0, 0\}$.

Missing Completely At Random

- Data are said to be MCAR if the failure to observe a value **does not depend on any data, either observed or missing**.
- The missing values of Z are MCAR if the probability of observing Z is independent of the values of Z that are observed or would have been observed.
- Under MCAR, **the observed data are just a random sample of all the data.**

Missing Completely At Random

- A trivial example would be, suppose that in a survey, each survey respondent decides whether to answer a question on earnings by rolling a die and refusing to answer if a "6" shows up.
- Formally,

$$\boxed{Pr(\boldsymbol{R}|Z) = Pr(\boldsymbol{R}|Z^{obs}, Z^{mis}) = Pr(\boldsymbol{R}|\boldsymbol{\theta})}$$

  Where $\boldsymbol{\theta}$ represents the set of parameters that govern the probability distribution of the missing indicator vector, R.
- In this trivial example, every respondent has a 1 in 6 chance of having missing data on earnings (regardless of any of their other data values or the missing earnings variable itself).

Missing At Random

- The data are said to be MAR if, conditional on the observed data, the failure to observe a value **does not depend on the data that are unobserved**. However, the conditional probability of missingness may depend on any observed data.
- At first glance, the term ”Missing At Random” may seem confusing, since missingness can actually be predicted (but is random after controlling for missingness due to observed quantities).

Missing At Random

- The missing values of $\mathbf{Z}$ are MAR if, conditional on the observed data, the probability of observing $\mathbf{Z}$ is independent of the values of $\mathbf{Z}$ that would have been observed.
- Note that this probability is not necessarily independent of the observed values of $\mathbf{Z}$.
  - Recall that $\mathbf{Z}$ includes Y and $\mathbf{X}$
  - So we could have, for example, that higher levels of $\mathbf{X}$ predict higher levels of Y and also increase the probability of not being observed. Then the values of Y that are observed are **not** a random sample of all values of Y in the population.

Missing At Random

- E.g. In a survey on earnings, gender, race, education, and age are recorded for all people in the survey. We find that the proportion of subjects missing the earnings variable varies by gender, race, education, and age, but conditional on those variables, we believe that missingness on the earnings variable is random.
- MAR is a more realistic assumption than MCAR, but usually adjustments must be made in the analysis because the subjects with fully observed data are no longer a random sample of all subjects.

Missing At Random

- Formally,

$$\boxed{Pr(R|Z) = Pr(R|Z^{obs}, Z^{mis}) = Pr(R|Z^{obs}, \boldsymbol{\theta})}$$

- Note that if the data are MCAR, then they are by definition MAR (i.e. MCAR is a special case of MAR). MCAR is a special case of MAR because under MCAR:

$$R \perp\!\!\!\perp Z^{obs} \implies Pr(R|Z^{obs}, \boldsymbol{\theta}) = Pr(R|\boldsymbol{\theta})$$

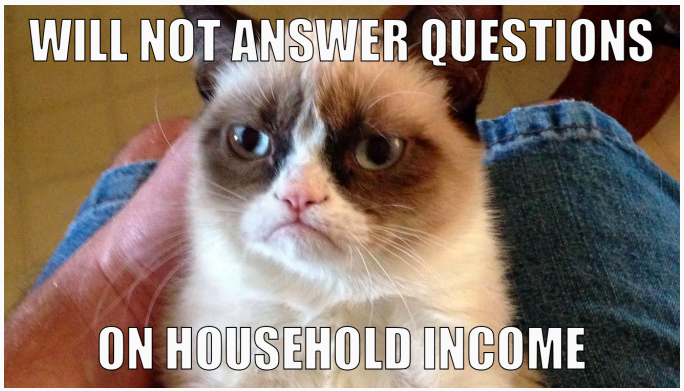
- Note also that we can test whether missingness follows MCAR or MAR in a given dataset by examining the data.

Missing Not At Random

- The missing data mechanism is said to be Missing Not At Random (MNAR) if the failure to observe a value **depends on the value that would have been observed**.
- The missing values of Z are MNAR if, conditional on the observed data, the probability that Z is missing depends on the missing values of Z.
- This can occur because missingness depends on the missing value itself or because missingness depends on unobserved predictors (which in turn affect the missing value).

Missing Not At Random

- For example, suppose in a survey on earnings that people with higher earnings are less likely to reveal them. Here, the probability of missingness on earnings varies with the value of the missing variable itself.
- Alternatively, suppose that "surly" people are less likely to respond to the earnings question, that "surliness" is predictive of earnings, and "surliness" is unobserved.



- Or suppose that people with college degrees are less likely to reveal their earnings, having a college degree is predictive of earnings, and there is also some non-response to the education question. Then the earnings variable is Not Missing At Random.

Missing Not At Random

- Here, the expression for the probability of missingness cannot be further simplified.

$$P(\mathsf{R}|\mathsf{Z}) = P(\mathsf{R}|\mathsf{Z}^{obs}, \mathsf{Z}^{mis})$$

  Valid inferences generally require specifying the correct model for the missing-data mechanism, distributional assumptions for the missing Z, or both. The resulting estimators and tests are typically sensitive to these assumptions.

- **We cannot use our empirical data alone to determine whether the data are MAR or MNAR.**[10]

[10] An empirical distinction between MNAR vs. MAR would require access to data which *were not observed.*

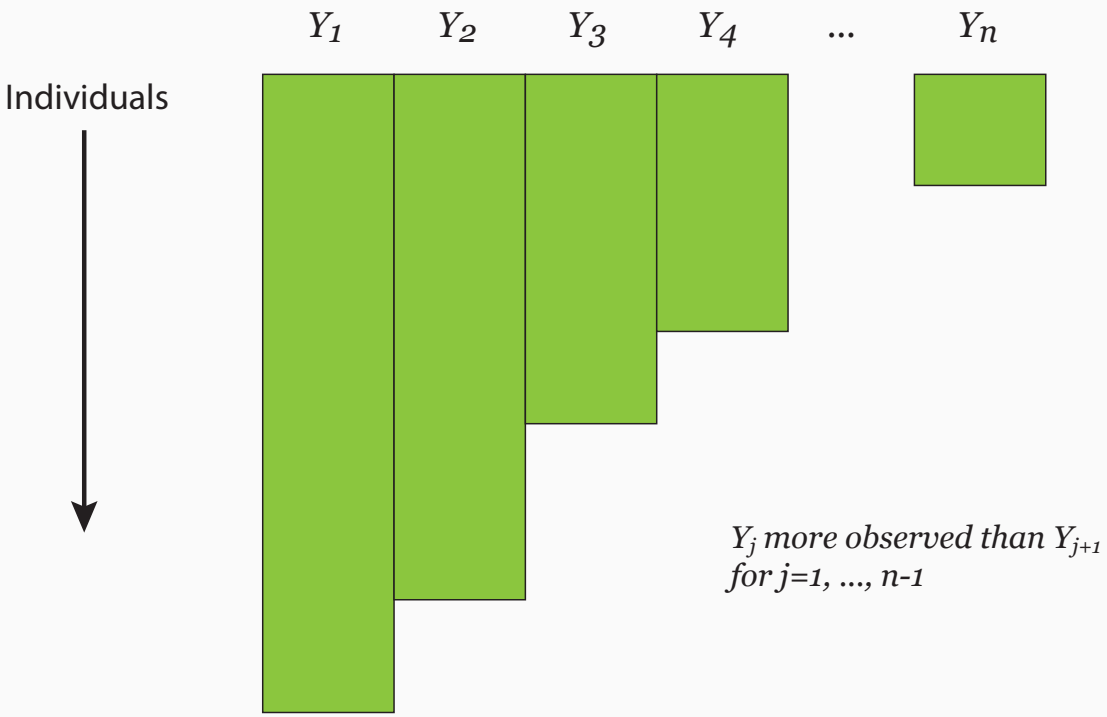
Patterns of missing data

Another important concept regarding missing data, particularly where there are multiple variables with missing values, relates to the **pattern** of missing data. If the data matrix can be rearranged in such a way that there is a hierarchy of missingness, so that observing a particular variable X_b for a subject implies that X_a is observed, for $a < b$, then the missingness is said to be **monotone**.

This has implications for some of the procedures for missing data imputation.

Dropout

Monotonic patterns of missingness are particularly relevant for longitudinal data since this is the pattern of missingness we expect to see with dropout or attrition.



Unit vs. Item Non-Response

In longitudinal settings, it is also useful to distinguish between **unit** non-response vs. **item** non-response.

- **unit non-response:** If subject i is not measured at time t, all of subject i's responses are missing at time t.
- **item non-response:** If subject i is measured at time t, but fails to answer one of the questions, then only that item is missing at time t.

In general, we expect to see both kinds of missingness in longitudinal datasets.

MCAR/MAR/MNAR: Additional comments

- It is tempting to think that the missingness affecting a variable always fits neatly into one of the three categories.
- From a practical viewpoint, the reality is more complex.
  - MNAR can take different forms and depart from MAR by different amounts.
  - It depends on what else has been observed!
- If enough additional information is collected, MNAR missingness can be converted (close) to MAR
  - Often used as justification for methods which assume MAR.
- However, the MAR or MNAR classification will **still be an assumption.**
  - Remember that it is not possible to distinguish between MAR and MNAR from the observed data.
  - Sensitivity analyses may be required.

Missing data handling methods

Methods to Address Missing Data

Several methods can be used to **address missing data**, including:

- Complete Case Analysis
- Ad Hoc Procedures (e.g., last observation carried forward (LCOF))
- Multiple Imputation
- Weighting and Model-Based Approaches

Whether a given method (e.g., complete case analysis) leads to bias will depend on whether the **proposed estimator** provides an **unbiased estimate** of the **estimand of interest** *under a given set of assumptions*.

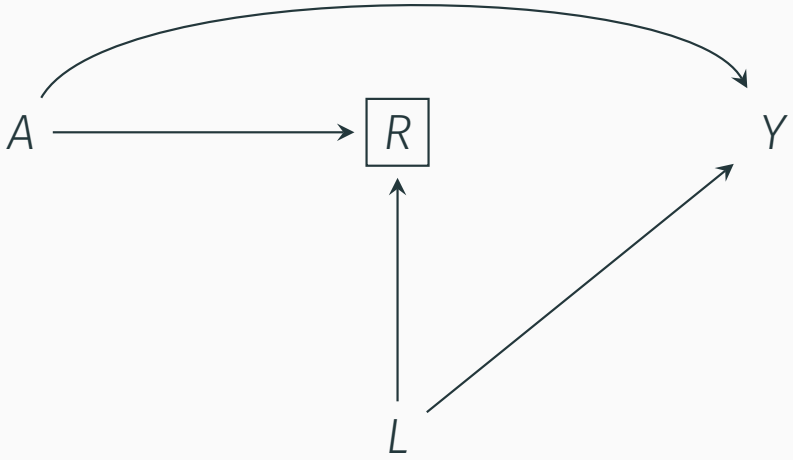
Complete Case Analysis

- missing data are ignored and only complete cases analyzed
- many computer packages do this by default
- **Advantages**: simple
- **Disadvantages**
  - often introduces bias
  - inefficient as data from incomplete cases discarded
  - the amount of discarded data can be potentially large if there are several variables that exhibit missingness.

Complete Case Analysis

- A complete case analysis may lose efficiency, but no bias is introduced when the data are MCAR.
- **In many if not most MAR scenarios, a complete case analysis will be both inefficient and biased.**
  - In data that are MAR, if missingness depends only on $\mathbf{X}$ and not on Y, then a complete case analysis will lead to unbiased estimates.
  - However, if the missingness depends on Y (and not necessarily on $\mathbf{X}$), then a complete case analysis will result in biased estimates.

Complete Case Analysis As Selection



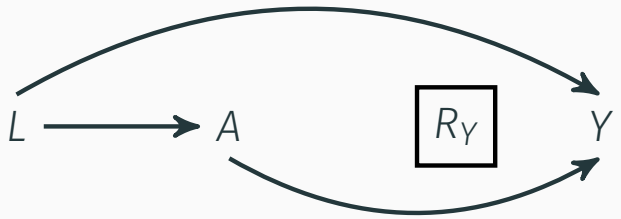
A **complete case analysis** is restricted to the stratum where $R = 1$ (observed cases). That is, the structural conclusions we discussed earlier apply when selection is specifically due to missing data.

Given the structure above, under what conditions can we recover an unbiased estimate of the effect of A on Y?

Structural Representations of Missing Data: MCAR Example

For example, consider the following DAG where we are interested in estimating the causal effect of A on Y. In the DAG below, we are missing data on the outcome Y.

We have $Z^{obs} = \{L, A\}$, and $Z^{mis} = \{Y\}$

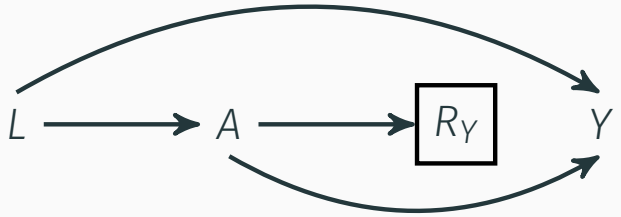


Data on Y are MCAR because $R_Y \perp\!\!\!\perp (Y, A, L) \implies R_Y \perp\!\!\!\perp (Z^{mis}, Z^{obs})$.[11]

[11] It can be helpful to subscript the missing indicator R by the variable(s) which are missing (e.g., R_Y if Y is the only variable with missing values). We will use this notation in the following DAGs.

Structural Representations of Missing Data: MAR Example

Now let's consider an alternative DAG where we assume that missingness in the outcome Y depends on the exposure A. Here, $Z^{obs} = \{L, A\}$ and $Z^{mis} = \{Y\}$:

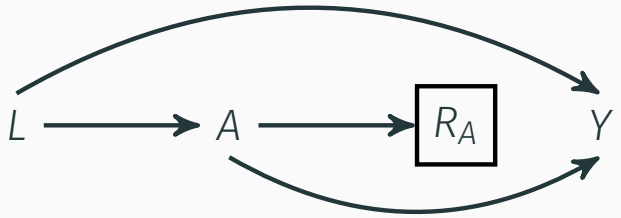


Data on Y are MAR because $R_Y \perp\!\!\!\perp Y | (A, L) \implies R_Y \perp\!\!\!\perp Z^{mis} | Z^{obs}$.

If the effect of A on Y varies across levels of L, then this is an example of selection bias without colliders!

Structural Representations of Missing Data: MNAR Example

Finally, let's consider a DAG where we assume that there is missing data for the variable A and missingness depends on A itself. Now we have $Z^{obs} = \{Y, L\}$ and $Z^{mis} = \{A\}$:

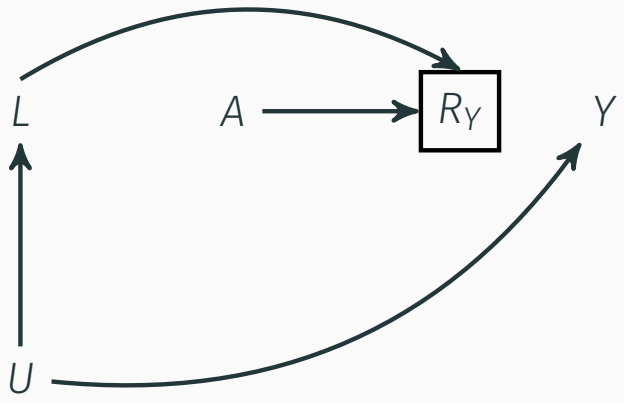


Data on A are MNAR because $R_A \not\perp\!\!\!\perp A|(Y, L) \implies R_A \not\perp\!\!\!\perp Z^{mis}|Z^{obs}$. We will return to this MNAR example after reviewing methods which can be used to address missing data.

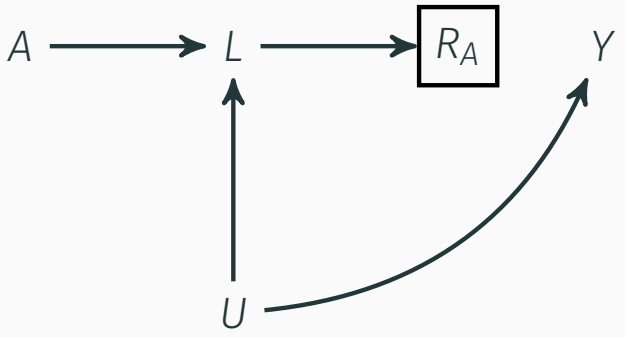
Structural Representations of Missing Data

Question: What kinds of missingness are represented by these DAGs?

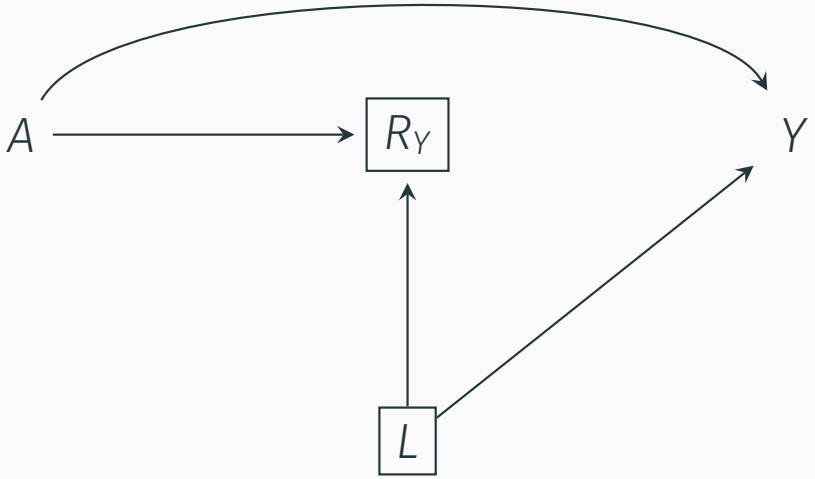
a)



b)



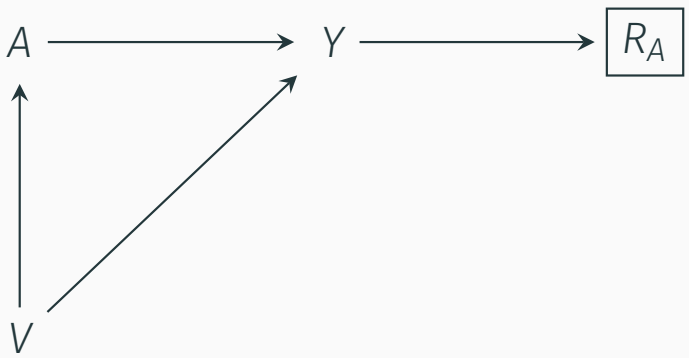
Example: Complete Case Analysis Under MAR



Under MAR, the probability of being observed can depend on observed variables (here it depends on A and L). In this simple situation, unbiased estimate of the effect of A on Y can be obtained in a complete case analysis by conditioning on L (e.g. via regression adjustment).

Note that conditioning in a complete case analysis will not solve all MAR situations; it depends on what variables are unobserved!

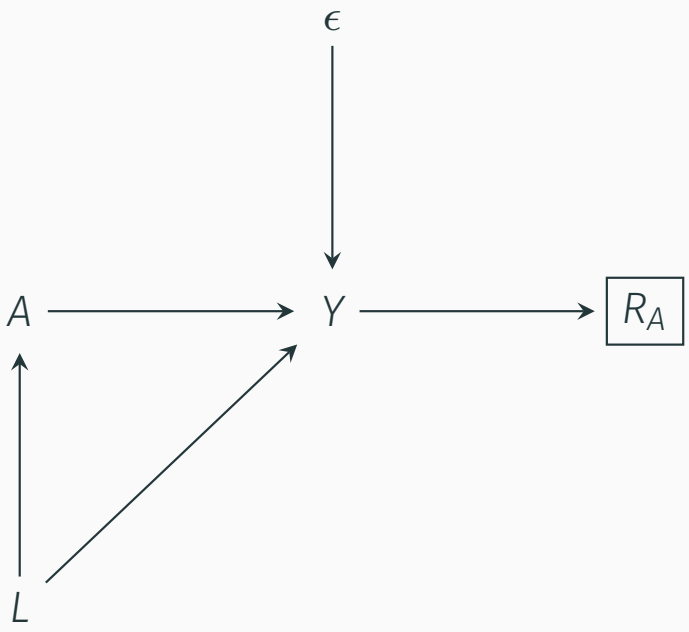
Example: Complete Case Analysis Under MAR



Consider the situation where the exposure A is partially observed and the probability of being observed depends on the outcome Y. We are interested in estimating the effect of A on Y.

The data are MAR, but it is not immediately obvious from this DAG why regression-based estimates of the effect of A on Y will be biased.

Example: Complete Case Analysis Under MAR



- What if I now tell you that Y is continuous and we will use linear regression to model $\mathbb{E}[Y|A, L] = \beta_0 + \beta_1 A + \beta_2 L$?
- Let's explicitly depict the random error term ϵ as a "cause" of Y. How does conditioning on R_A affect the correlation between A and ϵ?
- What effect does this have on estimation of $\hat{\beta}_1$ using OLS?

Example: Complete Case Analysis Under MAR

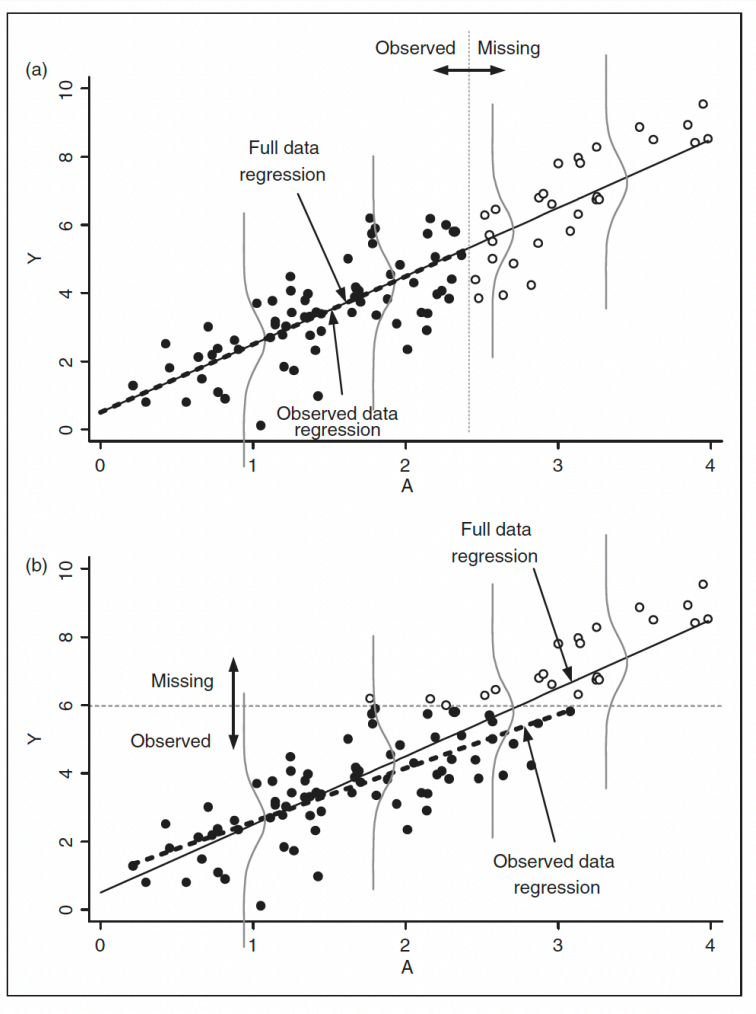
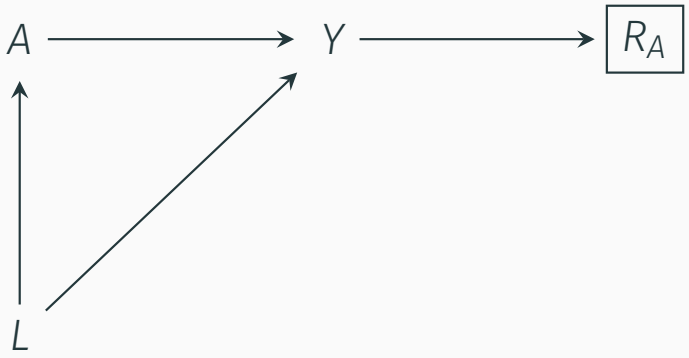


Figure 4 from Daniels et al. 2011: A comparison of (i) the full data estimate and (ii) the complete records estimate of the causal effect of *A* on *Y* under two different missingness mechanisms: missingness depends on *A* in panel (a) and missing depends on *Y* in panel (b)

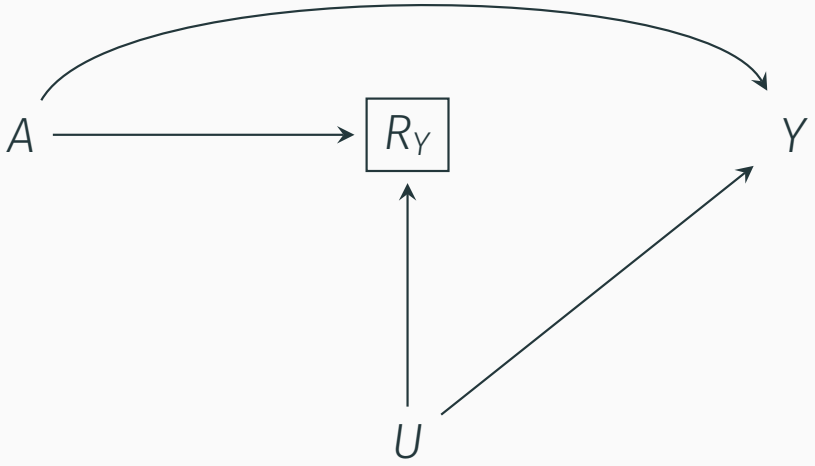
[11] Daniels RM, Kenward MG, Cousens SN, DeStavola BL. Using causal diagrams to guide analysis in missing data problems. *Statistical Methods in Medical Research* 2011;21(3):243-256.

Example: Complete Case Analysis Under MAR



- What if I now tell you that Y is dichotomous and we will use logistic regression to model $\mathrm{logit}(Pr[Y = 1|A, L]) = \beta_0 + \beta_1 A + \beta_2 L$?

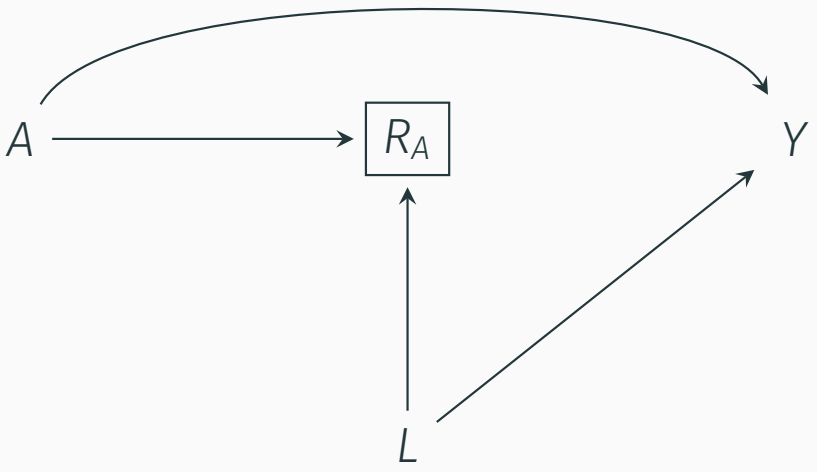
Example: Complete Case Analysis Under MNAR



Here, missingness in the outcome Y depends on U, which is unobserved.

This is an example of Missing Not At Random.

Example: Complete Case Analysis Under MNAR



Here, missingness in the exposure A depends on the values of A itself.

This is another example of Missing Not At Random.

But... note that by conditioning on L we can obtain an unbiased estimate of the association of A and Y!

Some take-home messages

- DAGs can help us build our intuition about how restricting an analysis to complete cases may bias our results in the presence of missing data.
- Whether or not there will be bias depends on the missingness mechanism, the estimand of interest, and the estimator used. We saw examples today where:
  - The data are MAR, but conditioning on predictors of selection can yield an unbiased estimate of an effect of interest in the observed sample.
  - The data are MAR, but restriction to complete cases induces bias when using a particular estimator.
  - The data are NMAR and we are unable to condition on anything in the complete case analysis that can help us recover an unbiased estimate of the effect of interest.
  - The data are NMAR, but conditioning on an observed predictor of selection suffices to yield an unbiased estimate of the effect of interest.

Complete Case Analysis

In a **complete case analysis,** only those observations for which *all variables are measured* are analyzed. This method is simple but **inefficient**. Complete case analyses **may also be biased**:

Data Are:	Does Complete Case Analysis Induce Bias?
MCAR	No
MAR	It depends. In general, if missingness depends on exposure A and/or measured covariates L but <u>not</u> on the outcome Y $\implies$ unbiased estimates. If missingness depends on the outcome Y $\implies$ biased estimates. In time-varying settings, complete case analyses will also often result in biased estimates.
MNAR	It depends. If missingness depends on a variable U which was not measured at all $\implies$ biased estimates. In <u>some</u> cases, complete case analyses may provide biased estimates for the marginal causal effect of interest in the target population but unbiased estimates for the conditional causal effect in the selected population. In the absence of effect modification on the scale of interest, an unbiased conditional causal effect would be equal to the marginal causal effect.

Complete Case Analysis

The potential for bias in a complete case analysis also **depends on the outcome model.** For example, complete case analyses are more robust to bias under logistic outcome regression:[12]

Variable(s) Missingness Depends On:	Linear Regression	Logistic Regression
None (i.e., data are MCAR)	Unbiased	Unbiased
Outcome	Biased[a]	Unbiased
Exposure (and possibly confounders)	Unbiased	Unbiased
Outcome and confounders	Biased	Unbiased
Outcome and exposure (and possibly confounders)	Biased	Biased[b]

[12] Adapted from Hughes RA, Heron J, Sterne JAC, Tilling K. Accounting for missing data in statistical analyses: multiple imputation is not always the answer. Int J Epidemiol. 2019 Aug 1;48(4):1294-1304.
[a] Biased in general, except when in truth there is no association between the outcome and the exposure.
[b] Biased in general, except when missingness depends on the outcome and exposure independently.

Ad Hoc Procedures

Procedure	Description	Major Disadvantages
Available cases	For each analysis, include all cases where the variable(s) involved in that analysis are present	Sample base may change across different analyses
Drop covariates	Completely drop covariate(s) with large amounts of missing data	May result in uncontrolled confounding
Missing indicator	Create a missing value indicator which is included in the analytic model(s)	Makes very strong assumptions; biased even under MCAR
Single imputation	"Fill in" each missing value with a single imputed value, then analyze the imputed dataset using standard methods	Ignores uncertainty in the imputed missing values; may be biased even under MCAR (e.g., LOCF)

Available case analysis (e.g. pairwise deletion)

- Available case methods include all cases where the variable(s) of interest is (are) present. This is often done in exploratory analyses of datasets where simple univariate summaries are generated based on the set of non-missing values for each variable. Similarly, correlation matrices may be computed based on fully observed data for pairs of variables.
- A **disadvantage** of this is that the sample base changes from variable to variable. Consequently, the resulting correlation matrices can be misleading. (Pairwise deletion is not recommended).

Dropping covariates

Another ad hoc approach involves dropping variables from the analysis that have a large proportion of missing values.

This is not attractive since it may lead to the exclusion of important confounders from the regression model, with consequent bias for estimating the effects of interest.

Single imputation

- Aim to ”fill in” missing values to create a single ”completed” (rectangular) dataset with imputed values, which is then analyzed using standard methods.
- Types of single imputation include
  - mean imputation
  - last observation carried forward (LOCF) (in longitudinal settings)
- Computationally simple but makes strong assumptions about missingness mechanism (usually MCAR in the case of mean imputation)
- ignores uncertainty about imputed missing values

Last observation carried forward (LOCF)

- widely used in clinical trial settings
- Assumption: all unseen measurements = last seen measurement
- is not even valid under the strong assumption of MCAR!

Missing indicator method

- For each variable with missing values, one creates a missing-value indicator to accompany the variable in all analyses. The missing value indicator takes the value 1 wherever the original variable is missing, 0 otherwise.
- For example, if X_1 is age at menarche and some subjects have X_1 missing, one creates a missing age-at-menarche indicator M_1 such that $M_1 = 1$ among subjects with X_1 missing, $M_1 = 0$ among the rest.
- One then replaces age at menarche in the regression with the missingness indicator, M_1, and the product of age at menarche and one minus the indicator, $X_1(1 - M_1)$ (note that this is zero if X_1 is missing).
- For categorical variables, this involves creating an additional category for missing values.

Missing indicator method

- Unfortunately, this method is biased under most conditions, **including under MCAR** (whereas the complete case analysis is unbiased under MCAR).
- Though the missing indicator has been discredited in the literature, it is still frequently used in practice. **We don't recommend that you use it!**
- See the simulation at the end of today's presentation to see why!

Simulation

To explore some of the implications of missing data in a simple, cross-sectional setting, I've taken an existing dataset and simulated missingness under MCAR and MAR so that we can compare the performance of

- complete case analysis
- missing indicator
- weighting

in estimating an exposure-outcome relationship.

In this dataset, we have measurements of $Y = log.FEV1$, $X_1 = height$, and $X_2 = age$ among 300 subjects at the Topeka site of the Six Cities Study (a study of the effects of air pollution on human health).

Complete case analysis under MCAR and MAR

Let's begin by comparing MCAR (where the probability of missing height does not depend on other variables) to MAR (where the probability of missing height depends on age and log.FEV1). Under both scenarios, I simulated 30% missingness.

- For MCAR, $P(R = 0) = 0.3$.
- For MAR,

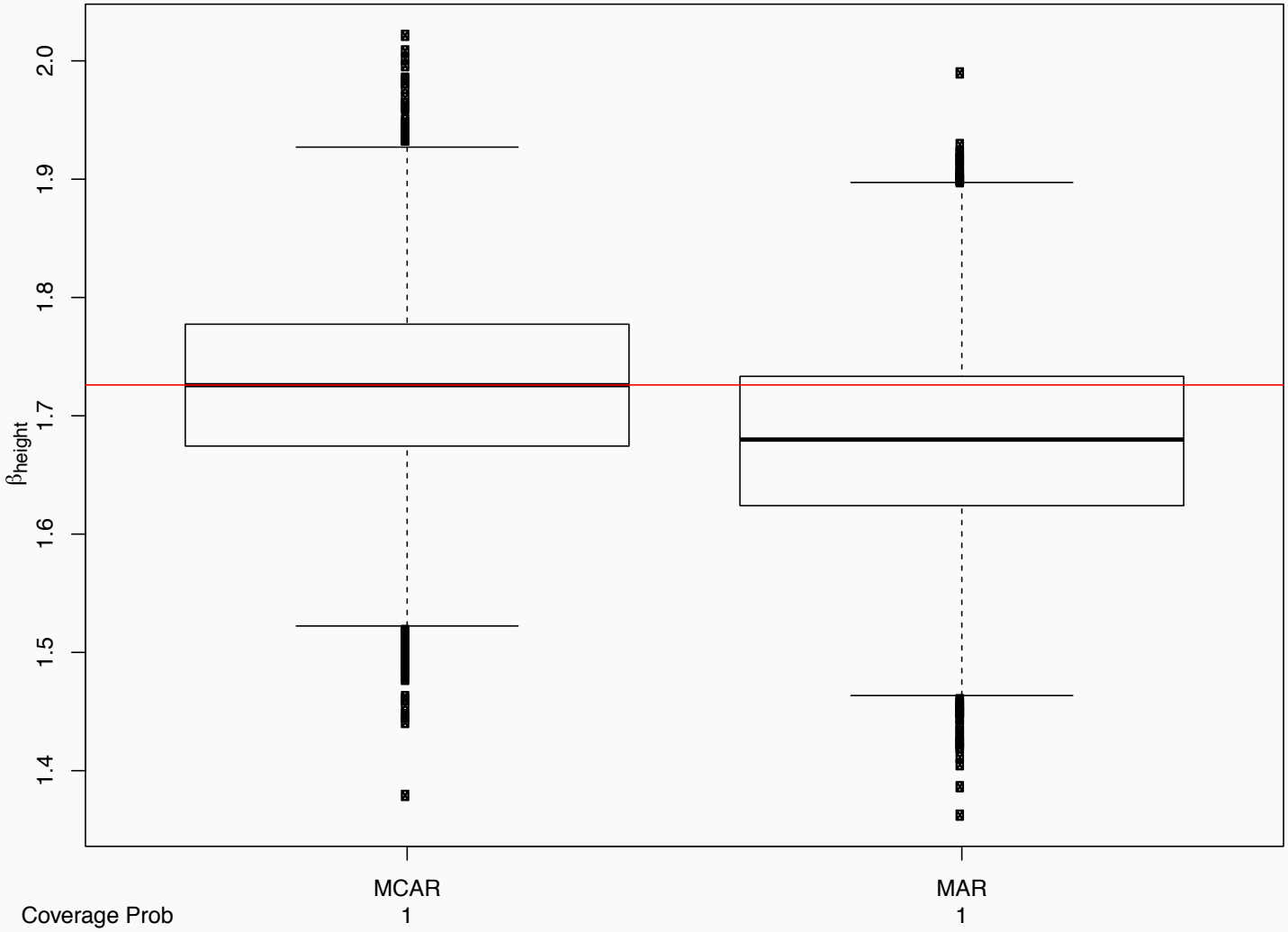
$$logit(P(R = 0|X_2, Y)) = \mathbf{log}(0.3/0.7) + 0.4 * (X_2 - \bar{X}_2) + 0.1 * (Y - \bar{Y})$$

I ran 5000 iterations of each simulation, estimating this linear regression model in each iteration:

$$Y_i = \hat{\beta}_0 + \hat{\beta}_{height} X_1 + \hat{\beta}_{age} X_2 + \epsilon_i$$

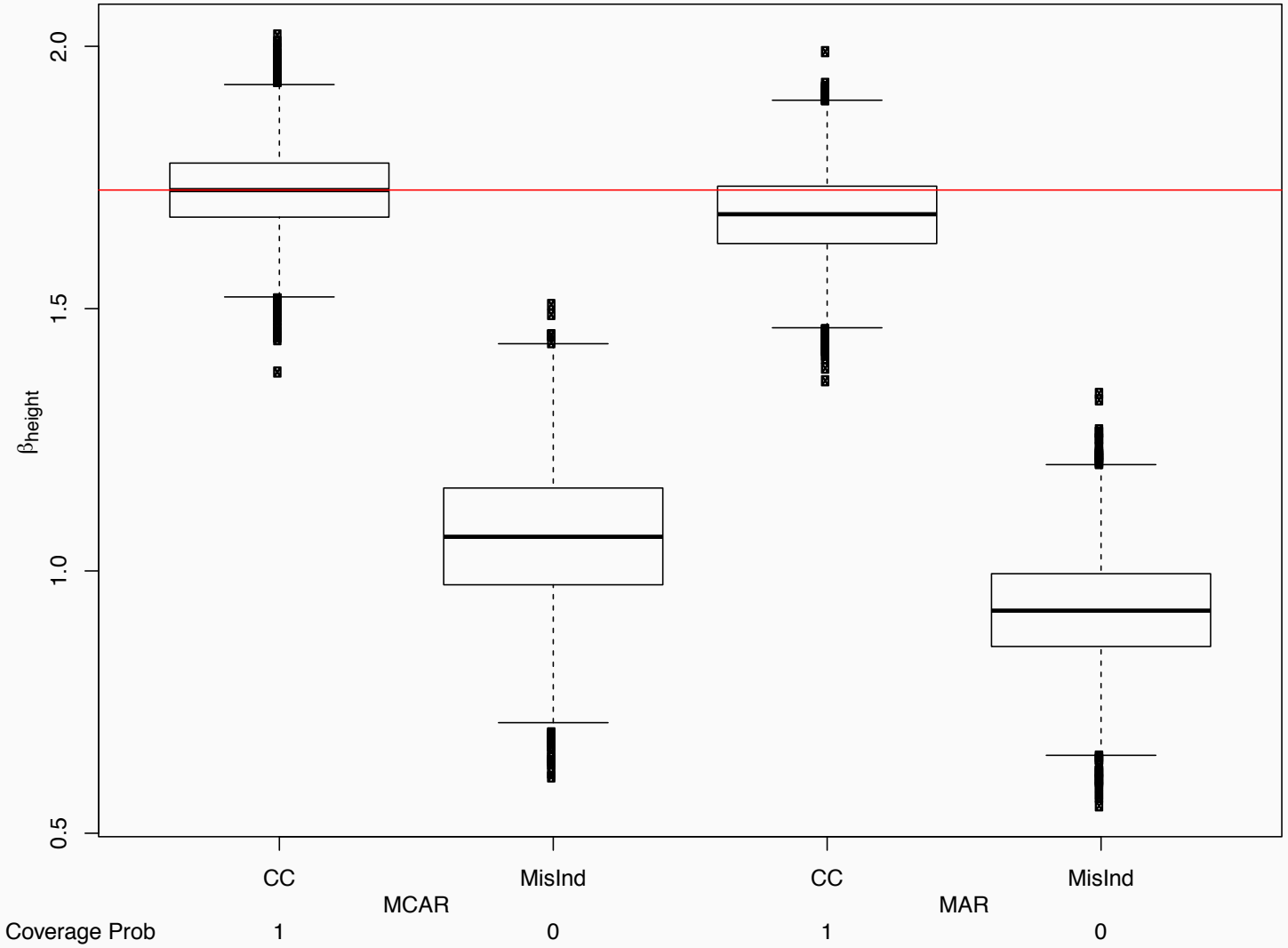
In each plot, I also summarize the coverage probability (proportion of times the true β_{height} was included in the confidence interval).

Complete case analysis under MCAR and MAR



Missing indicator analysis under MCAR and MAR

Now consider an analysis using the missing indicator method.

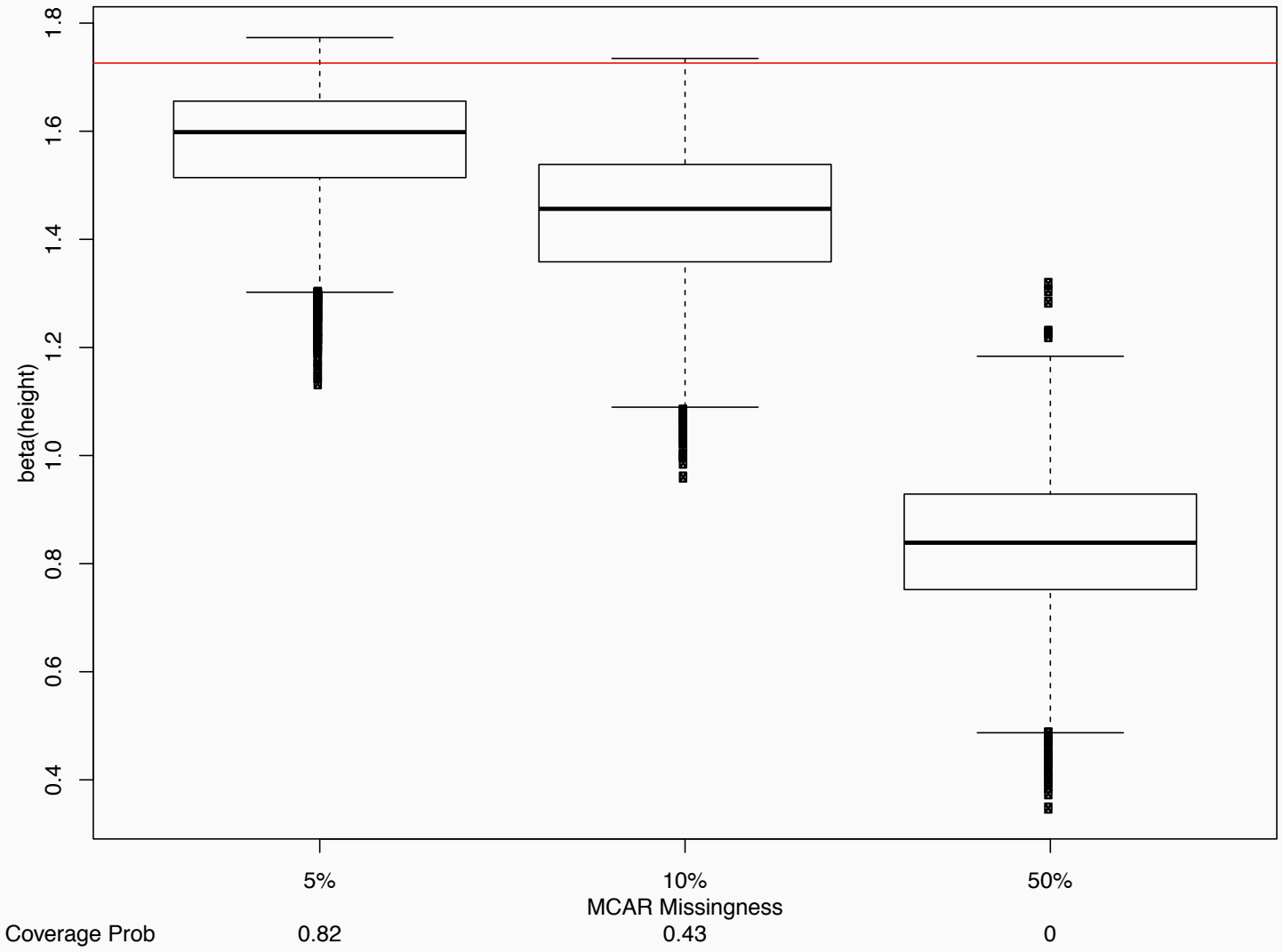


Missing indicator: MCAR scenarios

Now let's consider the missing indicator under the following MCAR missingness scenarios:

1. MCAR with 5% missing
2. MCAR with 10% missing
3. MCAR with 50% missing

Missing indicator: MCAR scenarios



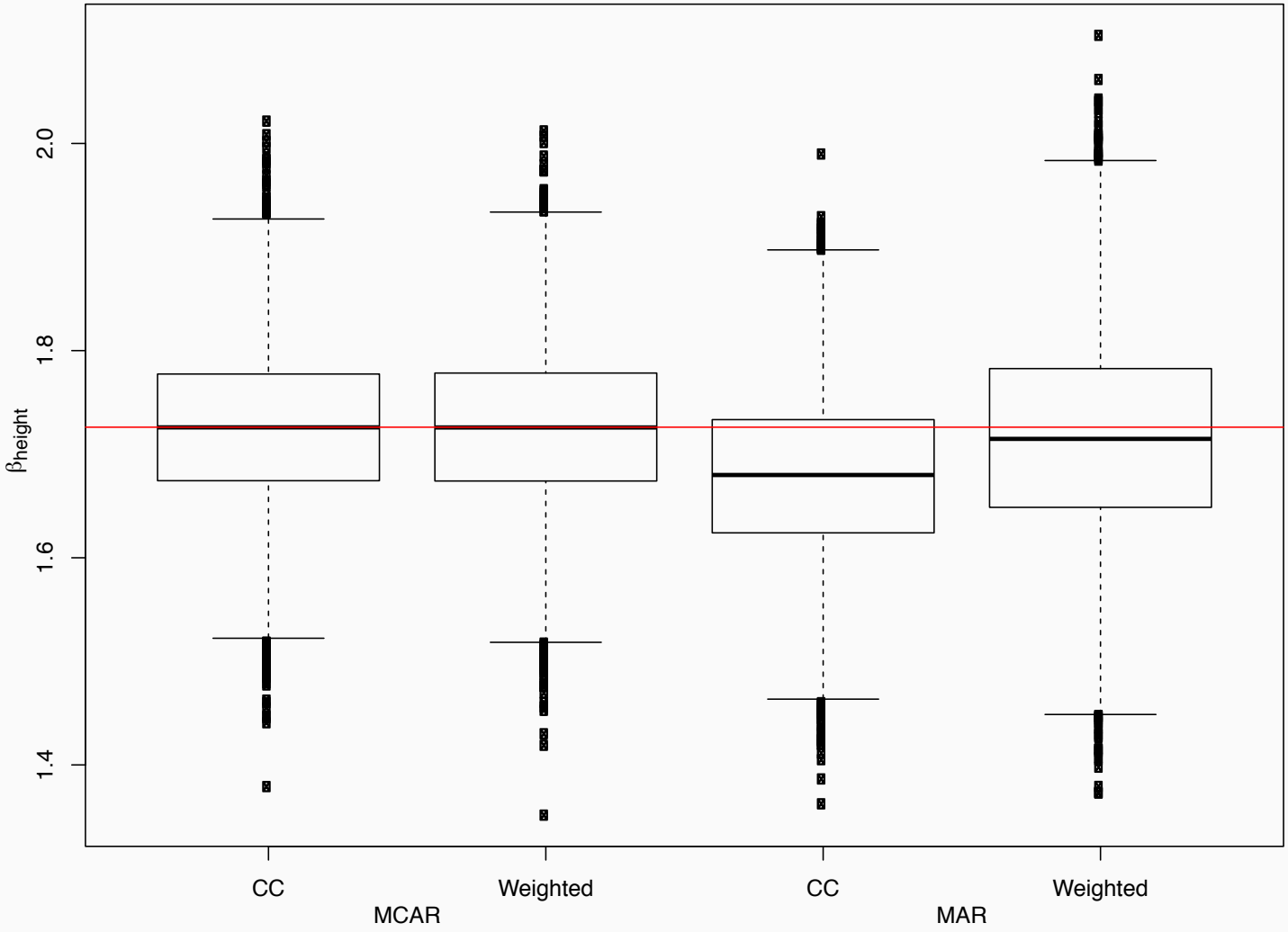
Weighted analysis: complete case vs. weighted under MAR

To compute the inverse probability weights, I fit a logistic regression model for

$$logit(P(R = 1)) = \theta_0 + \theta_1 age + \theta_2 log.FEV1$$

Note that this model included the fully observed Y variable as a predictor of missingness on height.

Weighted analysis: complete case vs. weighted under MAR



Simulation Summary

- Complete case analysis under MCAR gives an unbiased estimate of $\hat{\beta}_{height}$.
- Complete case analysis under MAR gives a biased estimate of $\hat{\beta}_{height}$.
- Weighted regression can recover an unbiased estimate of $\hat{\beta}_{height}$ under MAR.
- The missing indicator method yields biased estimates of $\hat{\beta}_{height}$ **even under MCAR!**

References

Two important textbooks:

1. Little RJA, Rubin DB. (1987) *Statistical Analysis with Missing Data*. New York: John Wiley Sons.
2. Schafer JL. (1997) *Analysis of Incomplete Multivariate Data*. Boca Raton: Chapman Hall/CRC.

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Some helpful papers:

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- Lu H, Cole SR, Howe CJ, Westreich D. Toward a Clearer Definition of Selection Bias When Estimating Causal Effects. *Epidemiology*. 2022 Sep 1;33(5):699-706. doi: 10.1097/EDE.0000000000001516. Epub 2022 Jun 6. PMID: 35700187; PMCID: PMC9378569.